

III rd Sem.-B.Tech. END-SEMESTER EXAMINATION, NOV.-DEC.,2023 Course

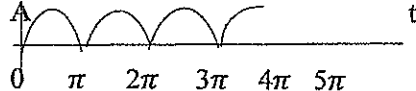
Code- EAEPC 05

Course Title- Signals and Systems

Time- 3 Hours

Max. Marks- 50

Note: Attempt all five questions. Missing data / information if any, may be suitably assumed & mentioned in the answer

Que	Attempt any two part of Questions BELOW	Marks	COs
1-a	Determine the power and Energy of the following signals: i) $x(t) = e^{-4t} u(t)$ ii) $y(n) = (\frac{1}{4})^n u(n)$ iii) $z(t) = \cos(t)$	5	CO 1
1-b	Determine if each of the following systems is invertible. If it is, construct the inverse system. If it is not, find two input signals to the system that have the same output. $y(t) = x(t-5)$ ii) $y[n] = \cos[x[n]]$	5	CO1
1-c	Explain advantages and disadvantages of discrete time systems over continuous time systems.	5	CO1
2-a	Input-output relationship for a system is given by $y[n] = x[n-2]x[n-4]$, determine whether the system is memoryless, invertible and linear.	5	CO2
2-b	A causal LTI system is characterized by the following difference Equation: $y[n] - (3/4)y[n-1] + (1/8)y[n-2] = 2x[n]$ Find out the impulse response.	5	CO2
2-c	Explain the properties of LTI system.	5	CO2
3-a	Find Fourier series of the following sequence- $f(t) = A \sin(\frac{\omega t}{2})$ 	5	CO3
3-b	Consider the rectangular pulse signal $X(t) = \begin{cases} 1 & t < T1 \\ 0 & t > T1 \end{cases}$ Find out its Fourier transform	5	CO3
3-c	Explain initial and final value theorem.	5	CO3
4-a	Consider a stable LTI system characterized by the differential equation- $\frac{d y(t)}{dt} + a y(t) = x(t)$, where $a > 0$; find out impulse response of the filter $h(t)$	5	CO4
4-b	Solve the differential equation given below using Laplace transform $\frac{d^2 i}{dt^2} + 4 \frac{di}{dt} + 5i = 5 u(t)$ if given initial conditions are $i(0) = 1$ and $\frac{di(0)}{dt} = 2$	5	CO4
4-c	Determine $x(t)$ for the following conditions: a) $x(t)$ is right-sided b) $x(t)$ is left-sided $X(s) = \frac{1}{(s+1)(s+2)}$	5	CO4
5-a	Find the inverse Z transform of the following signal $X(Z) = \log(1+z/a)$	5	CO5
5-b	Find the z transform as well as ROC for the following sequence (i) $(1/3)^n u(-n)$ and (ii) $(1/3)^n [u(-n) - u(n-8)]$	5	CO5
5-c	Explain Parseval's relation for discrete time periodic signals.	5	CO5